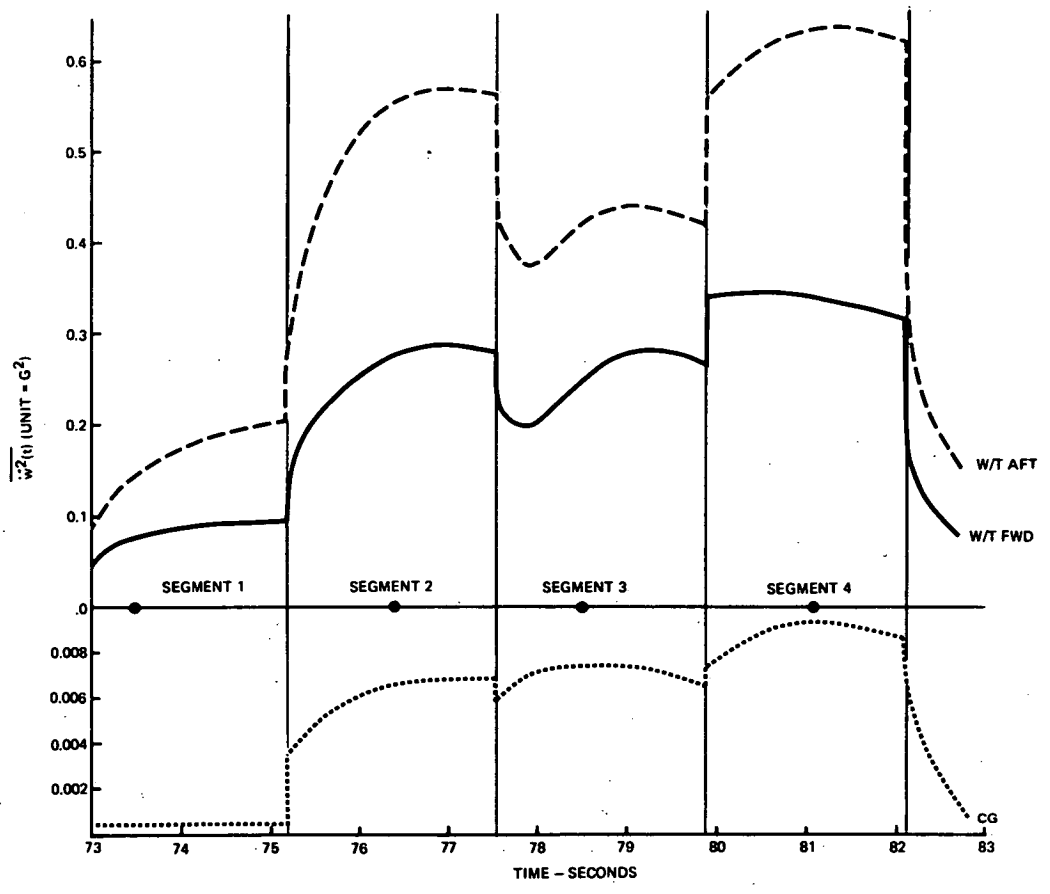

FIGURE 23. TIME VARYING ACCELERATIONS OF THE CG AND RIGHT-HAND WINGTIP DURING THE TRANSONIC MANEUVER, RUN 2, FLIGHT 871, $M_o = 0.925$, h = 10,668m, $\delta_n = 4°$, $\delta_f = 12°$

CONCLUSIONS

Based on the buffet flight tests and the related analytical processing, the following conclusions may be drawn:

1. Precise and detailed dynamic pressure data acquisition for aircraft during transonic buffet is feasible and can be productive. No significant adverse effect of the added instrumentation and change of airfoil geometry to the natural flow pattern was observed based on the dynamic flow development on the wing and the comparison with the tuft data. Fairly consistent mapping of the separation boundaries was achieved.

2. Shock instability in a steady-state uniform flow has been observed in previous wind tunnel and theoretical treatments under specific conditions depending on Reynolds number and local Mach number. The dynamic pressure of the F-5A showed definite oscillations where the separation boundary was located. It is uncertain whether the cause of the shock oscillation was solely due to the inherent shock instability; the cause might be partially attributable to the interaction with aircraft motion and angle of attack changes. This point remains to be resolved. In general, the amplitude and pattern of the shock oscillations were such that they were expected to contribute significantly to the aircraft motion.